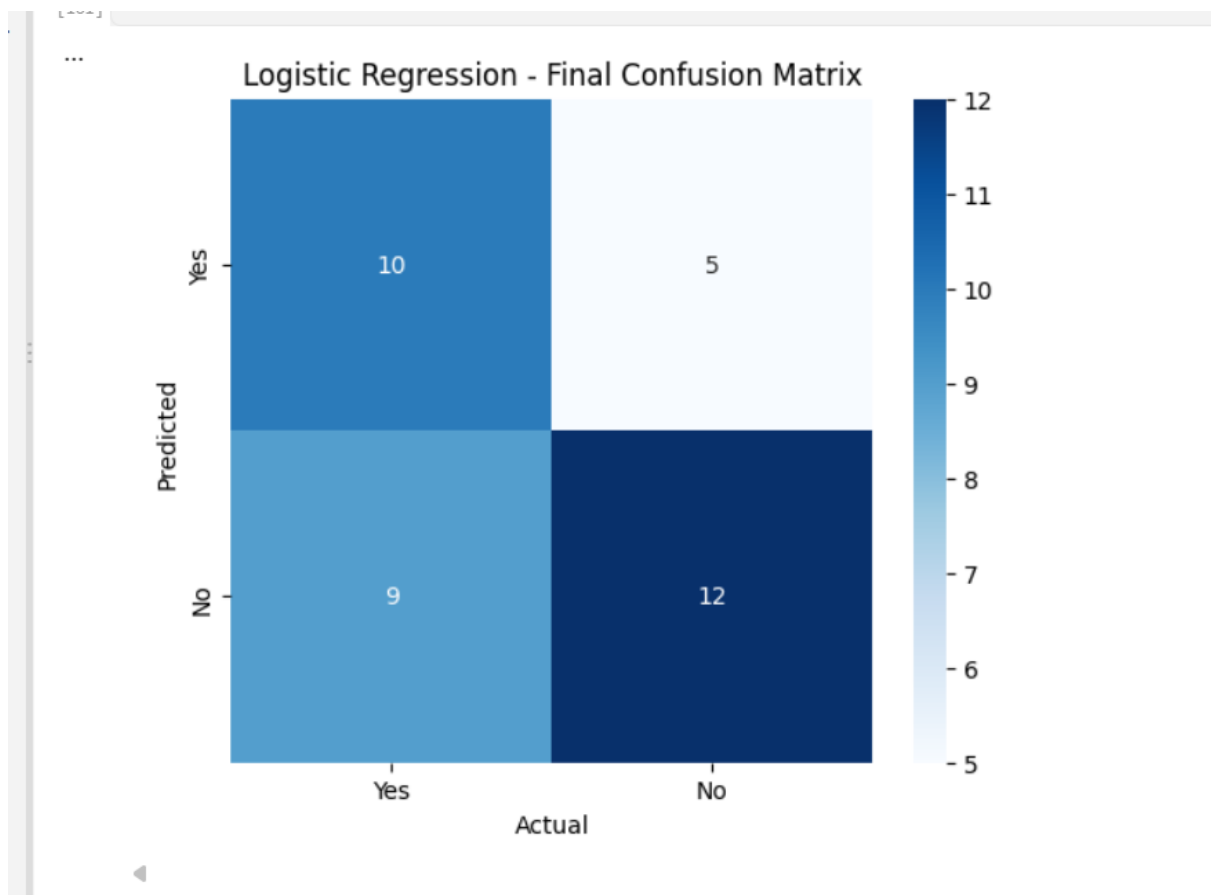


Model Comparison table:

	Model	Test accuracy	Test precision	Test recall	Test F1	Test Mean F1	CV-Test F1 Difference
0	Logistic Regression	0.6111	0.6667	0.5263	0.5882	0.650721	0.062521
1	KNN	0.4444	0.4667	0.3684	0.4118	0.602452	0.190652
2	Decision Tree	0.5000	0.5333	0.4211	0.4706	0.583803	0.113203
3	Random Forest	0.5556	0.6000	0.4737	0.5294	0.567434	0.038034

Confusion Matrix For my Model (Logistic Regression):



My Key Business Insights

Based on the exploratory analysis and model results, the following observations were identified:

Contract type is strongly associated with churn:

Monthly-contract customers had an observed churn rate of 62.50%, compared with 20.83% for 2-year contracts.

Customer satisfaction is associated with churn:

Higher satisfaction scores generally showed lower observed churn. Customers with a satisfaction score of 10 had a churn rate of 29.41%, compared with 69.23% at a score of 3.

Frequent support interactions are associated with higher churn:

Customers with 6 or more support calls in the last three months had an observed churn rate of 78.57%.

Higher data usage was associated with higher churn:

The highest data-usage quartile had an observed churn rate of 62.22%, compared with 44.44% in the lowest quartile.

The model can help identify potential churners:

Logistic Regression achieved a mean cross-validation F1-score of 65.07%. At the exploratory threshold of 0.30, recall increased to 73.68%, demonstrating that the model can identify more potential churners when the business prioritizes reducing missed churn cases.

Business Recommendations

Based on the observed churn patterns and model results, the following actions can be considered:

Focus retention efforts on monthly-contract customers:

Since monthly-contract customers showed a higher observed churn rate, the company can consider targeted retention offers, contract upgrade options, or incentives for longer-term plans.

Proactively engage customers with frequent support interactions:

Customers with 6 or more support calls showed substantially higher observed churn. These customers can be prioritized for service-quality reviews and proactive follow-up.

Monitor customers with lower satisfaction scores:

Lower satisfaction was generally associated with higher churn. Customer feedback, service recovery, and satisfaction-improvement initiatives can be used for customers showing signs of dissatisfaction.

Use churn predictions to prioritize retention activities:

The Logistic Regression model can be used as a screening tool to identify customers with a higher predicted probability of churn. A lower classification threshold can be considered when the business wants to identify more potential churners, while recognizing that this may generate more false-positive alerts.

Continuously evaluate the model with new customer data:

The current dataset is relatively small, so the model should be monitored and retrained as more customer data becomes available. Performance should also be validated on future unseen data before being used for automated decisions.

Limitations

The following limitations should be considered when interpreting the results:

Small dataset:

After removing exact duplicate records, the dataset contains 180 customer records. A larger dataset would provide more reliable estimates of model performance.

Data quality issues:

The original dataset contained missing values, inconsistent categorical labels, placeholders, and unusual numerical values. These were investigated and cleaned, but some unusual observations were retained where they were not clearly invalid.

Limited generalization:

The evaluation is based on one held-out test set and cross-validation on the training data. Performance on a different customer population may vary.

Preprocessing limitation:

Missing values were imputed before the train-test split during the cleaning stage. For a fully leakage-free production workflow, imputation should be performed inside the preprocessing pipeline using statistics learned only from the training data.

Threshold analysis limitation:

The classification threshold comparison was performed on the held-out test set for exploratory purposes. A final production threshold should be selected using validation data and business costs before deployment.

Association does not imply causation:

The EDA findings show relationships between customer characteristics and observed churn, but they do not establish that these factors directly cause customers to churn.

Further validation is required:

Before operational use, the model should be tested on a larger and more recent dataset and monitored over time.